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Evaluation of AR-HUD Interface During an Automated Intervention in Manual Driving

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Abstract— **Automated driving systems are envisioned as the future mode of transportation owing to their projected ability to reduce human error and achieve more efficient and comfortable transportation. Accordingly, designing an interface that ensures the situational awareness of the human operator to reduce confusion, false expectations, and over-reliance on the automated system is important. When a human operator is in control, the automated system is expected to handle troublesome situations that the human is unable to manage. Thus, an interface is required to provide the appropriate information when necessary so that the human operator can easily perceive the reason for the sudden automated intervention. In this study, such a scenario is highlighted, in which a simulated automated intervention avoided a potential collision with a pedestrian who suddenly appeared on the roadside. To convey the reason for the automated intervention, an augmented reality-based head-up display (AR-HUD) cue that targets the pedestrian is developed. To understand the effects of the AR-HUD cue on the speed at which a human operator can recognize a pedestrian and the contribution of this visual cue to the perception of acceptability and credibility of the automated intervention, we compared AR-HUD with a static head-up display (S-HUD) that displays a pedestrian symbol at the bottom portion of the windshield. The results showed that the AR-HUD cue yielded faster recognition of the targeted pedestrian and provided a relatively more acceptable perception of the automated intervention.**

I. INTRODUCTION

Automated cars are seen as the future of transportation, in which human error will be reduced, saving millions of lives, while providing a more energy efficient, comfortable, and convenient method of transportation. The National Highway Traffic Safety Administration defines six levels of car autonomy: Level 0 – human driver solely performs the driving; Level 1 – function-specific automation; Level 2 – simultaneous combined functions with automation, the current state-of-the-art; Level 3 – limited self-driving in which the driver needs to be ready to take over vehicle control at any time, Level 4 – full self-driving under certain circumstances; Level 5 – full self-driving in all circumstances in which humans are just passengers and need not be involved in driving [1]. Among these levels, Level 3 has a significant importance as it is expected to be utilized in the next generation of vehicles [2]. In recent years, several tech entrepreneurs have tested their automated driving systems in Level 3 to achieve the ultimate goal of accomplishing fully automated driving in the near future [3][4]. Unfortunately, in March 2018, a self-driving Uber car collided with a pedestrian during its test. The National Transportation Safety Board reported that one reason for the accident was that even though the system detected the crossing pedestrian with sufficient time before the collision, there was insufficient time for the vehicle operator to detect and react to the crossing pedestrian. This is a typical effect of automation complacency, and resulted in the failure of avoiding a collision [5]. This incident shows that the design and function of the human–machine interface (HMI) is crucial for ensuring that the human operator and automated system collaborate in a safe manner. A mutual understanding between the human operator and automated system ensures that the human operator obtains the appropriate amount of driving-related information at the right time, leading to the establishment of accurate situational awareness [6]. Such mutual understanding is essential for a suitable calibration of trust with acceptable and comfortable operation in general. A misunderstanding between the automated system and human operator regarding the action of the other party has the potential to cause false expectations as well as an over-reliance on the system capability. In contrast, if human operators have insufficient trust in the system capabilities, they may refuse to buy or use these systems.

”Automation surprise” is defined as the behavior of an automated system in which its action is inconsistent with the expectations of the driver [7]. Hurts et al. mentioned that automation surprise may occur before the operator becomes aware of its existence [8]. Automation surprise may also refer to a positive phenomenon in which a system takes control of a situation in such a way that the driver does not expect. However, even though the automation surprise occurs to handle an unexpected situation, it is important to minimize the automation surprise and reconcile the driver to the course of action of the automated system to maintain its acceptability and credibility. To mitigate the automation surprise, the reason for unexpected behavior should be provided by the system in a timely manner so that the operator can comprehend the situation and adjust his trust toward the system. Therefore, it is important to design a HMI that presents driving-related information corresponding to unexpected situations and the consequent behavior of the automated system in such a way that the human operator can promptly comprehend.

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(a) S-HUD cue appears at a distance of 50 m from a pedestrian standing on the left side of the road and it stays until the car passes the pedestrian

(b) AR-HUD cue appears at a distance of 50 m from a pedestrian standing on the left side of the road and it continues to move upward (0.5 m) and downward (0.5 m) until the car passes the pedestrian

Fig. 1. Design of (a)(S-HUD) and (b)(AR-HUD)

Drivers receive 80% of driving-related information visually while looking forward toward the road, which supports safe driving practices [9]. If the driving-related information matches the real environment that is in line with the gaze of the human operator, then the operator can recognize the information naturally and focus on driving without distraction. One way to accomplish this is the head-up display (HUD). HUDs have been proposed as a solution to reduce the frequency and duration that the eyes of the driver are off the road by projecting the visual cues of the driving-related information onto the windshield. When compared to head-down displays, HUDs have demonstrated reductions in the response times to unanticipated road events, resulting in smaller variances in lateral acceleration and steering wheel angle [10]. HUD-based interfaces provide intuitive information through appropriate static visual cues presented close to the outside view of the driver. However, they have demonstrated an increase in mental load, as indicated by the longer response times in high workload situations [11][12]. Another limitation of HUDs is the potential for clutter, which may interfere with the visualization of other objects in the panorama [13]. A pilot study showed that static visual cues increased the response times for detecting hazards owing to obstruction, possibly caused by the nonconformal nature of the static cue [14]. Furthermore, in a real environment, cueing of less conspicuous objects by using a static cue may not be easily noticed by the driver, potentially causing fatal injuries.

Previous research has identified that subjects noticed a target faster when it was explicitly pointed to when compared to a target that is not emphasized [15]. Herein, one strategy that can enhance the usage of HUD technology to present driving-related information is using augmented reality (AR) with HUDs. AR refers to the combination of real and artificial elements in the line of sight, with the aim of highlighting important objects to improve human performance [16]. The research showed that the detection accuracy of a target in an aviation visual search task was enhanced by cueing the target when the object was not conspicuous [17]. Moreover, the research demonstrated that AR can alert drivers and direct their attention to dangerous situations [18]. Thus, we believe that conveying driving-related information through visual cues using an AR-based HUD (AR-HUD) can enhance situational awareness and result in shorter response times for the driver when compared to the utilization of static HUD (S-HUD) cues.

Because the timing of the presentation of information reflects the system behavior and capability, it must be consistent with the expectation of the driver; otherwise, it may lead to a reduction in driver trust [19]. Previous research has shown the effectiveness of alarm promptness on trust [20]. While considering this, not only is the timing of information presentation critical, but the recognition timing of the driver pertaining to the information. However, there has been minimal investigation on recognition timing and its relation to acceptability and trust. In this study, we focused on such a scenario in which a simulated automated driving system suddenly intervenes in manual driving to avoid a potential collision with a pedestrian who unexpectedly appears on the roadside. To reduce the automation surprise and increase the acceptability of the sudden automated intervention, it is important to convey driving-related information to the driver in a timely manner. Reference [21] focused on four qualities that are important in a HMI for minimizing automation surprise: observability, predictability, directability, and timelessness. We believe that a well-designed HMI that cooperates with the AR-HUD design attributes would reflect these qualities. Accordingly, the reason for the unexpected maneuver of the automated system would be understood quicker, resulting in a more acceptable and credible perception. In this paper, we demonstrate a comparison between an AR-HUD cue and a S-HUD cue in such a scenario. The results of this comparison demonstrate that an AR-HUD based interface is better in providing prompt recognition. Correspondingly, it may be useful for a better acceptability and credibility assessment for automated intervention.

II. Method

A. Design of Augmented Reality-based Head-up Display (AR-HUD) and Static Head-up Display (S-HUD) Cues

Previous research utilized pedestrian pictograms [22][23] or master alerts [24][25] as S-HUD cues to increase the

Fig. 2. Driving simulator and the experimental environment

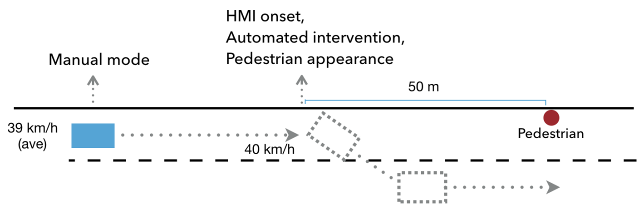

Fig. 3. Figure depicts the experimental scenario. The automated intervention occurs with the appearance of the pedestrian and the onset of the human–machine interface.

awareness of drivers regarding pedestrians. Research pertaining to AR-HUD utilized rectangles [23] or the rhombus shape of broken lines [26]. In our study, we designed S-HUD and AR-HUD cues to indicate a pedestrian on the road, which is the reason for the automated intervention. The S-HUD was designed using a pedestrian pictogram that showed the position of the pedestrian on the road (Fig. I(a)). Conversely, for the AR-HUD cue, we designed a cylinder with a 0.5 m radius that moves 0.5 m upward and downward by using the Blender software (version 2.79b). The AR-HUD cue is placed directly on the pedestrian (Fig. I(b)). The orange color is chosen for two cues (pedestrian pictogram and cylinder) to convey a warning rather than an emergency [27]. We placed the visual cues at the appropriate places on the screen using the driving simulation software settings.

B. Hypothesis

The mutual understanding between a human operator and an automated system is important in establishing an accurate situational awareness with a well-designed HMI such that the driver can comprehend the actions of the system, which may reduce the automation surprise and increase the willingness to use and trust the system. The design of such a HMI should include multiple modalities, such as visual, audio, and tactile cues. As the first step of this research, we focused on designing visual cues and exploring the objective and subjective responses as well as the impressions of the human operator toward the visual cues and automated intervention. We hypothesized that an AR-HUD cue would cause the driver to recognize the target that indicated a potential obstacle with a shorter response time than that with the S-HUD cue. Moreover, because the human operator would recognize the target faster, the reason for the automated intervention would be comprehended faster with the AR-HUD cue, resulting in a more acceptable and credible perception. We also hypothesized that the AR-HUD interface would provide a better user experience than the S-HUD interface owing to its explicit and dynamic design.

C. Experimental Setup

The UC-win/Road program (version 13) [1] was used as a simulation software along with five monitors, an adjustable driver seat, a steering wheel, a break, and an accelerator (Fig. 2). The maximum speed of the simulated car was set to 50 km/h. We examined two experimental conditions with the AR-HUD and S-HUD cues in the same driving route.

In the AR-HUD condition, the AR-HUD cue appeared on the pedestrian when the car was 50 m away. The cue was visible on the pedestrian until the car passed him. In the S-HUD condition, the S-HUD cue was located on the bottom side of the screen when the car was 50 m away from the pedestrian, and the cue was visible until the car passed the pedestrian.

The route was 5 km in the experimental session; moreover, the participants were asked to drive at a speed of approximately 40 km/h. The participants started driving the car in manual mode for 4.5 km, and the automated intervention scenario occurred at a distance of 0.5 km before the end of the route. At this point, a pedestrian suddenly appeared on the left side of the road; simultaneously, the simulated car, which was in manual mode initially, intervened automatically, and the manual mode was changed to the automated mode and the car changed lanes to the right 50 m before the pedestrian (Fig. 3). Simultaneously, with the appearance of the pedestrian and the onset of the automated intervention, the HMI cue was displayed. The rest of the trip was completed in the automated mode at a speed of 40 km/h. Both before and during the intervention process, no other information was presented to the subjects regarding the purpose of the intervention.

During the experiment, we set the time of the scenario to be before sunset, at which time the visibility of the objects was relatively low. Along the road, we also placed a group of pedestrians on the sidewalk and a construction scene on the right side of the road to distract the driver from the targeted pedestrian. Thus, we prevented a potential ceiling effect that may result from an easily observed target for both the aforementioned conditions. During the experiment, to track and evaluate the visual attention of the participants and their recognition time of the targeted pedestrian, we used Tobii Pro Glasses 2 [2] eye tracker.

(a) Example from the AR-HUD condition shows that the participant recognized the pedestrian in the second step of their gaze

(b) Example from the S-HUD condition shows that the participant recognized the pedestrian in the sixth step of their gaze

Fig. 4. Figures depict two examples pertaining to the difference between the gaze shifts of two participants in AR-HUD and S-HUD conditions

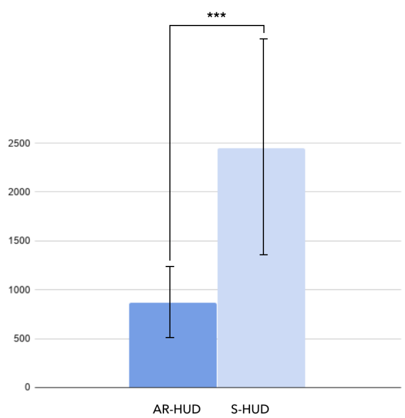

Fig. 5. Comparison of the recognition time of participants in the AR-HUD and S-HUD conditions *(***: p<0.001)*

D. Procedure

Our experiment was conducted on 36 participants (10 females, 26 males) ranging from 21 to 54 years (mean (M) = 34.638, standard deviation (SD) = 11.293). We conducted a between-subjects study to avoid the influence of learning effects. All participants possessed a current driving license. The participants were randomly divided into two groups of 19 and 17 members to which the AR-HUD and S-HUD conditions were applied, respectively. Upon arrival, the participants were given a brief explanation regarding the experiment. Before starting the experimental sessions, the participants underwent a 5 min practice session to accustom them to the driving simulator. The practice session was conducted on a different road than the road considered in the experiment. During the practice session, the participants used the simulated car manually and were exposed to the automated intervention. After the practice session, the experimental session was initiated.

After the experimental session, the participants were asked to complete three questionnaires: the Acceptance Questionnaire for evaluating the automated intervention with the AR-HUD- or S-HUD-based visual cue, Source Credibility Measurement [20] for evaluating the automated intervention, and User Experience Questionnaire for evaluating the visual cue [21].

The experiments were performed after obtaining approval from the Nagoya University Ethics Committee.

III. Results

Data from two participants and one participant in the AR-HUD and S-HUD conditions, respectively, were omitted from the objective and subjective data analyses owing to low data accuracy. In our analysis, we defined the "response time" as the time interval between the onset of the HMI and the first observation by the participants of the pedestrian who suddenly appeared by the roadside.

The average time of finishing the experimental conditions was 6 min 46 s (SD = 1 min 0 s). Additionally, the average speed of the participants just before the automated intervention and onset of the HMI was 39.218 km/h (SD = 8.142).

A. Eye Gaze Data Analysis

The analysis showed that the gazes of two participants in the AR-HUD conditions were already on the area of the targeted pedestrian before the pedestrian and visual cue appeared. We included the response times of these two participants by reformulating them as the average value of the shortest three response times of the other participants. Moreover, the gazes of three participants in the S-HUD condition were not aligned with the pedestrian at all. Therefore, the response times of these three participants were reformulated as the average value of the three longest response times from the rest of the participants. After these modifications, we conducted a two-sample t-test. The results showed that in the AR-HUD case, the attention of the participants shifted onto the pedestrian significantly faster at the $p < 0.05$ level than in the S-HUD case after the onset of HMI. *($t(15) = 4.825$, $p = 0.0001$).*

When compared to the S-HUD condition, the ratings for the AR-HUD condition pertaining to Q4 and Q13 were significantly higher ($t(31) = 1.708$, $p = 0.048$ and $t(31) = 2.753$, $p = 0.004$, respectively). The gazing pattern of the participants showed that under the AR-HUD condition, the

[1]http://www.forum8.co.jp/
[2]https://www.tobiipro.com/

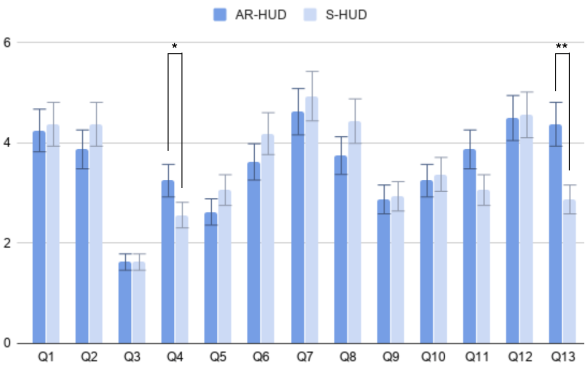

Fig. 6. Acceptance Questionnaire results show significant differences for Q4 and Q13. *(***: p<0.001, **: p<0.01, *: p<0.05)*

TABLE I

ACCEPTANCE QUESTIONNAIRE ITEMS THREE ITEMS ARE BASED ON HEAD-UP DISPLAY-BASED CUE, TWO ITEMS ARE BASED ON AUTOMATED INTERVENTION, AND SIX ITEMS ARE BASED ON THE COMPATIBILITY OF AUTOMATED INTERVENTION AND HEAD-UP DISPLAY-BASED CUE.

Code	Question
Q1	I could easily recognize the visual cue.
Q2	There is nothing wrong with the visual cue.
Q3	The visual cue was annoying.
Q4	Nothing was wrong with the execution of automated intervention.
Q5	The function of automated intervention was troublesome.
Q6	The visual cue allowed me to understand the execution of the automated intervention.
Q7	The reason why the automated intervention was executed can be understood from the visual cue.
Q8	I do not have any complaints about the visual cue.
Q9	Thanks to the visual cue, I did not have any anxiety during the automated intervention.
Q10	Thanks to the visual cue, the function of the automated intervention was reliable.
Q11	I will use this visual cue and automated intervention in my car.
Q12	I think the visual cue and the automated intervention will be helpful for my driving.
Q13	I was able to identify the targeted pedestrian quickly.
Q14	If you have any comments or feedback, please describe freely.

participants recognized the targeted pedestrian in fewer steps after the onset of HMI (Fig. II-D(a)). However, in the S-HUD case, the gazing pattern showed that the participants noticed the targeted pedestrian after they gazed at the S-HUD cue and then, after several steps of gazing, they shifted their gaze to the targeted pedestrian (Fig. II-D(b)).

When considered together, these results suggest that the AR-HUD-based visual cue achieved shorter response times of the drivers in recognizing the targeted pedestrian on the road.

B. Subjective Evaluations

1) Acceptance of the Automated Intervention with the Human–machine Interface (HMI) Cue: The Acceptance Questionnaire for the automated intervention created by us consisted of 14 items, in which the last was an open question (Table. I). The questionnaire responses were scored on a seven-point Likert scale: 7 (strongly agree), 6 (moderately agree), 5 (agree), 4 (neutral), 3 (disagree), 2 (moderately disagree), and 1 (strongly disagree). We applied a two-sample t-test to reveal the differences in the results from the questionnaire items for both conditions. The results showed significant differences between the two conditions for Q4 and Q13 (Fig. 6).

TABLE II

CORRELATION BETWEEN THE RESPONSE TIME AND THE ACCEPTABILITY QUESTIONNAIRE ITEMS *(left)*, THE SOURCE CREDIBILITY MEASURE FACTORS *(right top)* AND USER EXPERIENCE QUESTIONNAIRE (*(right bottom)*) IN AR-HUD AND S-HUD CONDITIONS

Acceptability Questionnaire		
Code	**Response Time**	
	AR-HUD	**S-HUD**
Q1	-0.691**	0.3
Q2	-0.621*	-0.096
Q3	0.01	-0.354
Q4	-0.095	-0.058
Q5	0.363	0.085
Q6	-0.301	-0.242
Q7	-0.188	-0.589*
Q8	-0.344	0.072
Q9	-0.423	-0.007
Q10	-0.387	0.045
Q11	-0.411	0.064
Q12	-0.546*	0.008
Q13	-0.543*	0.166

Source Credibility M.		
Factor	**Response Time**	
	AR-HUD	**S-HUD**
Comp.	-0.531*	-0.285
Caring.	-0.564*	-0.548*
Trust.	-0.637**	-0.392

User Experience Q.		
Factor	Response Time	
	AR-HUD	**S-HUD**
Attr.	-0.400	0.359
Eff.	-0.564*	0.183
Pers.	-0.102	-0.197
Dep.	-0.257	0.349
Stim.	-0.291	0.169
Novelty	-0.081	0.010

We also checked the correlations between the Acceptance Questionnaire items and the response time values for the two conditions. For the AR-HUD condition, we identified significant correlations that were strongly negative *(r(16) = -0.691, p = 0.003)* and moderately negative *(r(16) = -0.621, p = 0.01; r(16) = -0.546, p = 0.028 and r(16) = -0.543, p = 0.029)* between the response time values and Q1, Q2, Q12 and Q13, respectively. For the S-HUD condition, we identified a significant moderately negative correlation between the response time and Q7 *(r(15) = -0.589, p = 0.016)* (Table II *(left)*).

2) Credibility of the Automated Intervention: The Source Credibility Measurement items were also scored on a seven-point Likert scale. The results of a two-sample t-test showed no statistically significant difference between the AR-HUD and S-HUD conditions pertaining to the factors of Competence, Caring, or Trustworthiness (Fig. 7) on the p <0.05 level. However, when we checked the correlations between the response time and the credibility factors for each condition, we observed significant moderately negative correlations between the response time and the factors of Competence *(r(17) = -0.531, p = 0.034)*, Caring/Goodwill *(r(17) = -0.564, p = 0.022)*, and Trustworthiness *(r(17) = -0.637, p = 0.007)* for the AR-HUD condition. In addition, we observed a significant moderately negative correlation between the response time and the Caring/Goodwill factor *(r(17) = -0.548, p = 0.027)* for the S-HUD condition (Table II *(right top)*).

3) User Experience Quality of the HMI Cue: The User Experience Questionnaire items were again scored on a

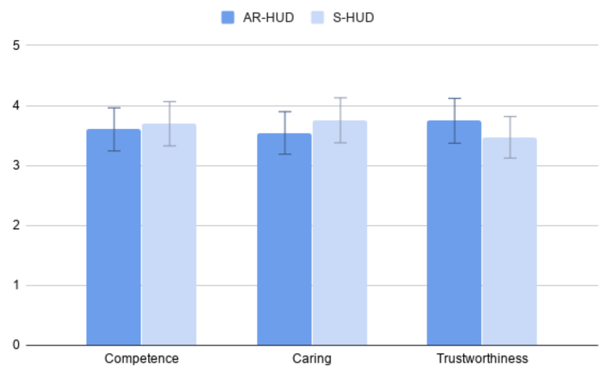

Fig. 7. Source credibility measurement results

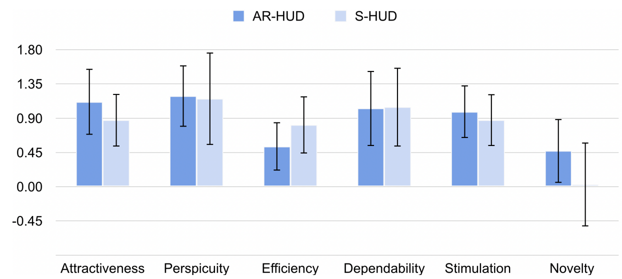

Fig. 8. User Experience Questionnaire results

seven-point Likert scale. The results of a two-sample t-test showed no statistically significant difference for the Attractiveness, Perspicuity, Efficiency, Dependability, Stimulation, or Novelty factors on the $p < 0.05$ level (Fig. 8). However, we observed a moderate negative correlation between response time and the Efficiency facfor ($r(17) = -0.564$, $p = 0.022$) for the AR-HUD condition (Table II *(right bottom)*).

4) User Comments and Feedback: Participants primarily complained about the difficulty in operating the steering. One participant commented that she could not pay sufficient attention to the road because she was concentrating on driving accurately. Predominantly, the comments of the participants were regarding the insufficiency of the warning system. Three participants from the AR-HUD group and one participant from the S-HUD group commented that they wished to have an audio warning along with the visual cue. Two participants from the AR-HUD group and four participants from the S-HUD group stated that they were surprised and scared by the sudden intervention of the automated system. Three participants indicated that they did not feel safe when the car changed lanes without providing any safety confirmation; consequently, they preferred the car to stop. One participant reported that the driving method of the automated system did not match the method of the participant. Furthermore, three participants indicated that they could not understand if they or the automated system was driving.

IV. Discussion

In this research, we compared the effects of the AR-HUD- and S-HUD-based visual cues on human operators upon an automated intervention to avoid a potential collision with a pedestrian who has suddenly appeared on the roadside. The results showed that the AR-HUD-based visual cues provided significantly shorter recognition time of the pedestrian and encouraged the human operators to demonstrate a marginally better perception of the automated intervention. While driving, evoking spatial attention allows drivers to selectively process visual information through the prioritization of an area within the visual field. Research has demonstrated that when spatial attention is evoked, an observer is typically faster and more accurate in detecting a target that appears in an expected location when compared to an unexpected location [28]. According to the ”spotlight metaphor”, the focus of attention is analogous to the beam of a spotlight [29]. The movable spotlight is directed at one location and everything within its beam is noted and processed preferentially, while information outside the beam remains unattended. Our findings of a faster recognition of the pedestrian by shifting the attention of the human operators to a specific location by utilizing an AR-HUD cue also support this theory. In addition, the results of the subjective evaluation (Q13) support our objective findings that human operators were aware that they could easily recognize the target. Furthermore, the indication that the gazing patterns of the participants demonstrated fewer shifts until the targeted pedestrian was noticed in the AR-HUD condition confirmed our hypothesis (Fig. II-D). Based on these findings, we can infer that the human operators can recognize the targets quicker when they are explicitly and dynamically indicated rather than implicitly and statically indicated on the windshield.

We also hypothesized that with a rapid recognition of the target, the acceptance of the automated intervention by the participants will be higher, and more credit is expected to be given to the automated system when an AR-HUD-based visual cue is used. The negative correlation between the recognition time and the three questions from the Acceptability Questionnaire (Q1, Q2, Q12 and Q13) in the AR-HUD condition showed that as the recognition time decreased, the perception of recognition of the target improved. These results can be closely related to the credibility perception of the automated system. The findings of recent research demonstrated that when a system is more transparent and provides the reasons for its behavior, then the users may become more forgiving and accepting toward the system, even if the behavior of the system is perceived to be bad [30][31]. Herein, even though the results of the credibility analysis did not show a meaningful difference between the AR-HUD and S-HUD conditions, we can infer from the correlation results that the perception of the human operators regarding the credibility of the automated system may depend on how fast they recognize the reason for the behavior of the system.

The automated intervention was generally perceived as abrupt and caused dissatisfaction to the participants for both conditions. Therefore, the considerably low ratings for Q9 may indicate this unpleasant perception of the automated system; moreover, unlike our expectations, the AR-HUD was not effective in enhancing this perception even though it provided faster recognition of the target. One reason for this might be the insufficient modality of the interface, as indicated by the comments of certain participants. As these types of abrupt interventions, in other words, automation surprises, may likely occur in the future with real autonomous cars in real-world scenarios, it is important to leverage the unpleasant perception of human operators through other modalities, such as audio or tactile cues, in addition to a HUD-based cue.

Limitations: This experiment was conducted using a driving simulator and two-dimensional HUD-based visual cues. The results of this study may change owing to the different depth perceptions of the AR and HUD technologies. In addition, the number of participants used in this study is relatively low. Nonetheless, our findings suggest that the two HUD-based cues merit further investigation. Future studies would benefit from larger sample sizes and a more realistic application of AR-HUD and S-HUDs, as our marginally modest experiment raised the possibility that some of the factors that were non-significant may have had certain effects that we were unable to detect.

V. CONCLUSIONS

In this paper, we demonstrated that an AR-HUD-based visual cue could provide a faster response time and facilitate the human operator recognizing a potential obstacle. The results also showed that with a shorter recognition time, the credibility of the automated system is likely to increase. While considering the design of a more acceptable and credible interface during an unexpected automated intervention, it was speculated that using only a visual cue may not provide a sufficient understanding between the system and the human operator. In future studies, we will examine different modalities and their combinations to support and satisfy human operators for better automated driving experiences.

REFERENCES

[1] N. H. T. S. Administration *et al.*, "Preliminary statement of policy concerning automated vehicles," *Washington, DC*, pp. 1–14, 2013.

[2] J. Bierstedt, A. Gooze, C. Gray, J. Peterman, L. Raykin, and J. Walters, "Effects of next-generation vehicles on travel demand and highway capacity," *FP Think Working Group*, vol. 8, pp. 10–1, 2014.

[3] T. Vanderbilt, "Let the robot drive: The autonomous car of the future is here," *Wired Magazine, Conde NAST, www. wired. com*, pp. 1–34, 2012.

[4] S. L. Poczter and L. M. Jankovic, "The google car: driving toward a better future?" *Journal of Business Case Studies (JBCS)*, vol. 10, no. 1, pp. 7–14, 2014.

[5] "National transportation safety board. collision between vehicle controlled by developmental automated driving system and pedestrian," https://www.ntsb.gov/news/events/Documents/2019-HWY18MH010-BMG-abstract.pdf, november 19, 2019.

[6] M. R. Endsley, "Toward a theory of situation awareness in dynamic systems," *Human factors*, vol. 37, no. 1, pp. 32–64, 1995.

[7] S. Dekker, *Report of the flight crew human factors investigation conducted for the Dutch safety board into the accident of TK1951, Boeing 737-800 near Amsterdam Schiphol Airport, February 25, 2009*. Lund University, School of Aviation, 2009.

[8] K. Hurts and R. J. de Boer, "Automation surprise looked at from a demands-resources perspective," in *Proceedings of the human factors and ergonomics society Europe chapter annual conference*, 2015, pp. 221–233.

[9] A. S. Cohen and R. Hirsig, *Feed forward programming of car drivers' eye movement behavior: A system theoretical approach*. Swiss Federal Institute of Technology, Department of Behavioral Science, 1980.

[10] Y.-C. Liu, "Effects of using head-up display in automobile context on attention demand and driving performance," *Displays*, vol. 24, no. 4-5, pp. 157–165, 2003.

[11] C. D. Wickens, R. Martin-Emerson, and I. Larish, "Attentional tunneling and the head-up display," 1993.

[12] E. Fischer and R. F. Haines, "Cognitive issues in head-up displays," 1980.

[13] S. Y. Cheng, A. Doshi, and M. M. Trivedi, "Active heads-up display based speed compliance aid for driver assistance: A novel interface and comparative experimental studies," in *2007 IEEE Intelligent Vehicles Symposium*. IEEE, 2007, pp. 594–599.

[14] M. C. Schall and P. Gavin, "Attraction without distraction: Effects of augmented reality cues on driver hazard perception," 2010.

[15] R. T. Azuma, "A survey of augmented reality," *Presence: Teleoperators & Virtual Environments*, vol. 6, no. 4, pp. 355–385, 1997.

[16] D. L. Fisher and K. C. Tan, "Visual displays: The highlighting paradox," *Human factors*, vol. 31, no. 1, pp. 17–30, 1989.

[17] M. Yeh and C. D. Wickens, "Display signaling in augmented reality: Effects of cue reliability and image realism on attention allocation and trust calibration," *Human Factors*, vol. 43, no. 3, pp. 355–365, 2001.

[18] M. Tonnis, C. Sandor, G. Klinker, C. Lange, and H. Bubb, "Experimental evaluation of an augmented reality visualization for directing a car driver's attention," in *Fourth IEEE and ACM International Symposium on Mixed and Augmented Reality (ISMAR'05)*. IEEE, 2005, pp. 56–59.

[19] A. M. Parkes and S. Franzen, *Driving future vehicles*. CRC Press, 1993.

[20] G. Abe and J. Richardson, "Alarm timing, trust and driver expectation for forward collision warning systems," *Applied ergonomics*, vol. 37, no. 5, pp. 577–586, 2006.

[21] O. Carsten and M. H. Martens, "How can humans understand their automated cars? hmi principles, problems and solutions," *Cognition, Technology & Work*, vol. 21, no. 1, pp. 3–20, 2019.

[22] A. Wuryandari, Y. Gondokaryono, and I. M. Y. Widnyana, "Design and implementation of driver main computer and head up display on smart car," *Procedia Technology*, vol. 11, pp. 1041–1047, 2013.

[23] M. T. Phan, I. Thouvenin, and V. Frémont, "Enhancing the driver awareness of pedestrian using augmented reality cues," in *2016 IEEE 19th International Conference on Intelligent Transportation Systems (ITSC)*. IEEE, 2016, pp. 1298–1304.

[24] D. Large, V. Banks, C. Harvey, and G. Burnett, "Towards a predictive model of driver acceptance of active collision avoidance systems," 2018.

[25] C. Merenda, H. Kim, J. L. Gabbard, S. Leong, D. R. Large, and G. Burnett, "Did you see me? assessing perceptual vs. real driving gains across multi-modal pedestrian alert systems," in *Proceedings ofx the 9th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*, 2017, pp. 40–49.

[26] M. L. Rusch, M. C. Schall Jr, P. Gavin, J. D. Lee, J. D. Dawson, S. Vecera, and M. Rizzo, "Directing driver attention with augmented reality cues," *Transportation research part F: traffic psychology and behaviour*, vol. 16, pp. 127–137, 2013.

[27] A. Chapanis, "Hazards associated with three signal words and four colours on warning signs," *Ergonomics*, vol. 37, no. 2, pp. 265–275, 1994.

[28] M. I. Posner, "Orienting of attention," *Quarterly journal of experimental psychology*, vol. 32, no. 1, pp. 3–25, 1980.

[29] M. I. Posner, C. R. Snyder, and B. J. Davidson, "Attention and the detection of signals." *Journal of experimental psychology: General*, vol. 109, no. 2, p. 160, 1980.

[30] N. Tintarev and J. Masthoff, "A survey of explanations in recommender systems," in *2007 IEEE 23rd international conference on data engineering workshop*. IEEE, 2007, pp. 801–810.

[31] R. Sinha and K. Swearingen, "The role of transparency in recommender systems," in *CHI'02 extended abstracts on Human factors in computing systems*, 2002, pp. 830–831.